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Using diverse local optima for setting kernel parameters in support vector regression: Forecasting emerging market credit spreads

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ABSTRACT

We propose a novel approach for determining support vector regression (SVR) kernel parameters in the presence of multiple local optima. In contrast to existing approaches focusing on identifying a single 'best' tuning parameter setting, an impractical goal in many financial market applications, our framework employs a global optimization algorithm to produce a collection of competitive SVR kernel parameter candidates, and applies the model confidence set test to select the most accurate subset from the collection of promising candidates. We use our approach to predict credit spreads for four mature emerging market sovereign borrowers. Combining our forecasts into simple forecast combinations and contrasting them against random forests, standard SVR, and conventional benchmark model forecasts, we find that forecast combinations from our most accurate model sets deliver substantial gains in forecast accuracy. Furthermore, our analysis of the forecasting results provides useful insights into credit risk pricing by international investors.

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1. Introduction

Support vector regression (SVR) techniques pioneered by Vapnik (1995) have been widely applied in forecasting applications in engineering, bio-informatics, and decision sciences; see for example Lins et al. (2012), Zhang and O'Donnell (2020), and Bhatia (2019). The recent evidence highlighting their superior performance for predicting financial market series has been attributed to the greater generalization ability of the structural risk minimization principle they implement relative to the empirical risk minimization principle in standard linear and non-linear regression techniques and neural networks (Dai et al., 2013; Sermpinis, Stasinakis, Rosillo, et al., 2017). Despite these successes, SVR kernel parameter selection remains an essential but problematic aspect for the practical application of SVR techniques. This stumbling block emerges because in order to utilize SVR with a given kernel, the modeler must choose a set of kernel, or 'tuning', parameters that ultimately determine how well the SVR model will characterize the data. There is ample evidence that the choice of tuning parameters dramatically influences the predictive power of the model in the context of financial series modeling. In particular, Stasinakis et al. (2016), Law and Shawe-Taylor (2017), Sermpinis, Stasinakis, and Hassanniakalager (2017), and Sermpinis,

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Stasinakis, Rosillo, et al. (2017) emphasize that SVR forecasting performance is highly sensitive to the choice of tuning parameters, while pointing out that there is little formal guidance on how to set these parameters.

Lacking compelling theoretical guidance, financial modelers often resort to applying an optimization technique, with a grid search being the most frequently selected, as in Ding et al. (2008), Dai et al. (2013), and Dash et al. (2023). Alternative approaches include biologically inspired global optimization algorithms in Pai et al. (2006), Sermpinis et al. (2015), Stasinakis et al. (2016), and Sermpinis, Stasinakis, and Hassanniakalager (2017), and a Bayesian approach in Law and Shawe-Taylor (2017). A common assumption in each of these approaches is that they can identify, with some degree of certainty, a single 'best' set of tuning parameter values producing a single best SVR forecasting model for a given kernel. However, in many financial market applications, the existence of multiple local optima in the objective function employed for determining tuning parameters makes it impossible for even global optimization techniques to distinguish with certainty among a number of potentially 'best' solutions. In particular, in bond credit spreads and other interest-rate-related applications, the problem is linked to the strong persistence in credit spread and interest rate data series along with some macroeconomic variables employed as predictors; this is well documented, for example, by Ang and Piazzesi (2003), Kim (2009), Audzeyeva and Fuertes (2018), and Klavdívko and Rusý (2023). Kim (2009) emphasizes that high persistence in the data series results in some aspects of the model being difficult to capture, no matter how frequently the data are sampled, leading to imprecise parameter estimation. He further points out that using a flexible model specification like SVR can result in an optimization function containing multiple optima, each capturing various aspects of the data with differing emphasis. Consequently, even global optimization techniques can struggle to successfully identify a unique global solution among perhaps many plausible local alternatives. Furthermore, such alternative solutions likely reflect complementary information about the data that can be useful for forecasting but is overlooked when a single solution is chosen.

To address this issue, we propose a novel approach that–unlike existing approaches advocating the selection of one 'best' setting for kernel parameters of the predictive model–bypasses the challenge of finding a unique global optimum of the objective function and the single 'best' kernel parameter setting by allowing a chosen SVR kernel to generate multiple competitive forecasts, thus improving the robustness of SVR estimation. The multiplicity of forecasts arises because an SVR kernel can reasonably accommodate a collection of good choices for tuning parameters. Accordingly, we propose a strategy for managing such multiplicity. First, our approach employs a global optimization algorithm to generate a set of promising model candidates, each employing a plausible tuning parameter set. In the next step, we select a relatively small set of the most accurate models from the collection of promising candidates by applying the Model Confidence Set (MCS) test of Hansen et al. (2011) that identifies a subset of models with superior forecast accuracy. Finally, depending on the modeler's objectives, the subset of most accurate forecasts can be employed directly or can serve as inputs in forecast combinations.

Samuels and Sekkel (2017) previously proposed using the MCS test for selecting (trimming) model forecasts to improve the accuracy of forecast combinations. However, our application of the MCS test differs from these authors in that we employ the test for selecting the most accurate forecasts generated using a given model and kernel function pairing. Our goal to improve the robustness of SVR estimation so that it does not require the modeler to select a single tuning parameter setting is also distinct from theirs.

Our subsequent empirical analysis applies our SVR framework to forecasting credit spreads of emerging market economies (EMEs). To the best of our knowledge, this is the first study that applies the machine learning methodology in the context of *emerging* bond markets, with previous literature predominantly focusing on stock, bond, and derivative markets of *developed* economies, cryptocurrencies, and global commodities; see for example Law and Shawe-Taylor (2017), Stasinakis et al. (2016), Sermpinis, Stasinakis, Rosillo, et al. (2017), Risse (2019), Khashanah and Shao (2022), and Wei et al. (2023). Our analysis is of relevance to market practitioners, as EME bond markets have grown in importance, with EME bonds becoming a core global asset class, accounting for one-quarter (US$ 16 trillion) of the global sovereign bond market by the end of 2021 (Onen et al., 2023). As such, they play a major role in portfolio allocation and the risk management decisions of international investors, as well as in maintaining global financial stability by financial market regulators and policymakers.[2]

The scant literature focusing on EME credit spread out-of-sample (OOS) predictability has so far relied on traditional prediction methods underpinned by the assumption of linearly formed investor expectations. In particular, Audzeyeva and Fuertes (2018) show that linear predictive models employing country term structure factors as predictors, known to produce competitive forecasts in the context of riskless debt (e.g. Diebold & Li, 2006), cannot beat a random walk for predicting emerging market bond credit spreads.[3] Furthermore, Wang et al. (2019), applying panel data regression estimation methods in the context of both developed and emerging market credit default swaps, struggle to identify a uniformly best estimator that can deliver superior forecasts. Thus, we extend the analysis in these studies by using highly flexible SVR model specifications, permitting more complex and realistic non-linear expectation mechanisms.

[2] Fekete and Janosik (2020) find that since the pandemic, emerging bond markets have provided more efficient price discovery of relative credit value than credit default swaps, used as an alternative market-based measure of sovereign credit risk. Yet, little is known about their predictability in real time, in sharp contrast to the voluminous literature on riskless developed market sovereign bonds (Ang & Piazzesi, 2003; Diebold & Li, 2006; Fama & Bliss, 1987; Ludvigson & Ng, 2009, among others).

[3] Audzeyeva and Fuertes (2018) further find that employing additional global and country-specific predictors, beyond the credit spread level, slope, and curvature factors, improves the predictive ability of models, but that even with these additional predictors, the model-based forecasts cannot always outperform a random walk.

Our analysis further adds to nascent literature on applying the SVR methodology to forecasting sovereign credit spreads, with Da Silva Fernandes et al. (2019) being among the few exceptions. We extend their analysis in several important directions. First, we expand the eurozone market-based evidence in their study to emerging markets dispersed across Asia, Europe, and Latin America. Second, our analysis of quarter-ahead predictability, relevant to longer-term institutional investors such as insurance companies and pension funds, complements the day-ahead and week-ahead predictability evidence in their study that is of more relevance to short-term investors such as speculators and arbitragers. Third, our focus on the credit spread predictability during a period of relative global market stability, characterized by more pronounced country-specific variations, augments their evidence relating to financial and debt crisis periods in the eurozone, marked by high market comovements. Finally, in contrast to these authors' forecasts utilizing data relating to the target variable alone, and aligned with our focus on longer-term predictability, we employ an extended predictor set that also exploits the information content in global and domestic macroeconomic predictors.

Consequently, we apply our framework to OOS forecasting of credit spreads for four relatively mature emerging markets–Brazil, Mexico, Turkey, and the Philippines–previously analyzed by Audzeyeva and Fuertes (2018) using conventional linear models. The four sovereigns have been constituents of the J.P. Morgan EMBI+ sovereign emerging market bond index, a market benchmark for large and liquid U.S.-dollar-denominated bonds since the early 2000s.[4] As the evidence to date on the relative performance of various SVR kernels for financial series has been inconclusive, we employ SVR with linear, radial basis function (RBF), and polynomial kernels to generate credit spread forecasts (e.g. Sermpinis, Stasinakis, Rosillo, et al., 2017).

We find that our approach generates a relatively small set of SVR forecasting models with superior predictive ability for each sovereign borrower. Our evidence further indicates that employing simple forecast combinations from the most accurate model sets delivers superior forecasts relative to random forests, standard SVR specifications, and conventional benchmarks from the bond credit spread forecasting literature. Moreover, we find that our approach differentiates the predictor sets it uses for borrowers with different dynamics in credit quality, providing useful insights into the pricing of sovereign credit risk by international investors.

The paper proceeds as follows. Section 2 provides a review of existing approaches to determining tuning parameters, while Section 3 outlines our forecasting framework. Section 4 describes the data and details of our empirical application. Section 5 discusses the empirical findings, and Section 6 concludes the paper.

[4] All four borrowers constitute large markets with a rich cross-section of traded bonds, permitting the estimation of the credit spread curve factors that serve as predictor variables in our analysis. Other sovereigns do not possess a sufficiently large cross-section of bonds throughout the sample period for the credit spread curve estimation.

2. Existing approaches to determining tuning parameters

In this section, we introduce the standard SVR methodology for which kernel specifications serve as inputs, followed by a review of current approaches to determining SVR kernel parameters.

2.1. Support vector regression

SVR implements structural risk minimization, thus minimizing an upper bound on the generalization error and enhancing the reliability of its OOS predictive performance over standard neural networks and linear regression among conventional estimation methods which minimize the error on observed data (Vapnik, 1995). Furthermore, SVR makes an attractive alternative to conventional methodologies like linear regression when forecasting financial time series, as it requires fewer assumptions about the data generating process. For example, it does not assume that residuals are normally distributed or homoscedastic. In addition, the various kernels available for SVR readily accommodate nonlinearities in the relationship characterizing the impact of various predictors while, at the same time, limiting the impact of overfitting. Also, because of the SVR algorithm's use of a so-called hinged loss function, it is likely less sensitive to the presence of outliers in the data (Kecman, 2005).

Using a training sample containing I observations of a scalar target variable y and a vector of predictor variables $x \in \mathcal{R}^Q$, SVR constructs predictive models for q-step-ahead forecasts of y of the form

$$y_{t+q} = f(x_t) + u_t = w_0 + \sum_{i=1}^{I} w_i k(x_i, x_t; \Upsilon) + u_t \tag{1}$$

Kernel function $k(x_i, x; \Upsilon)$ in Eq. (1) effectively maps the input data vector x into a higher-dimensional space representing a w_i-weighted linear sum of terms that can better predict the target variable y, due to its superior flexibility, with subscript t denoting a time-t value of a variable at hand. The vector of tuning parameters, Υ, provides a way for varying the nonlinear mapping of the data. This non-linear mapping, often referred to as the kernel trick, provides a variety of parsimoniously parameterized nonlinear functional forms (Hofmann et al., 2008). Table 1 presents three kernel functions frequently employed in financial time series modeling: a linear function, RBF, and polynomial function.

SVR chooses a vector of weights w_i that minimizes the regularized empirical risk function:

$$\begin{aligned} &\min \frac{1}{2}\|w\| + C\sum_{i=1}^{I}(\xi_i + \xi_i^*), \quad \text{such that} \\ &y_{i+q} - f(x_i) \le \epsilon + \xi_i, \;\; f(x_i) - y_{i+q} \le \epsilon + \xi_i^*, \\ &\qquad \xi_i, \xi_i^* \ge 0 \;\; \text{for all } i. \end{aligned} \tag{2}$$

Here, the bandwidth parameter, ϵ, determines an ϵ-insensitive region for characterizing empirical risk, and the regularization parameter, C, determines the tradeoff between

Table 1
Kernel functions.

Name	Kernel function $k(x_i, x)$	Kernel-specific parameters[a]
Linear	$(x_i{}^T x)$	–
Radial basis function[b]	$e^{-\frac{1}{\psi^2 \|x_i - x\|^2}}$	ψ
Polynomial[c]	$(\gamma(x_i{}^T x) + s)^g$	s, g, γ

[a] Each kernel function, Eq. (1), requires a choice of the regularization parameter, C, and the bandwidth parameter, ϵ, along with any kernel-specific parameters, altogether comprising the vector of tuning parameters Υ.
[b] Parameter ψ controls the radius of influence of individual observations: a larger ψ reduces the radius of influence.
[c] The polynomial kernel generalizes the linear kernel function by providing a nonlinear response to the dot product value: the larger the g, the more nonlinear the response. Here, γ moderates the sensitivity to the nonlinear interaction term, while s determines the location of the zero response.

a measure of flatness of the function, $\|w\|$, and the level of empirical risk.

The predictions of the parameterized function $f(x_i)$ can violate the constraint, but at a cost proportional to C. With this so-called double-hinged loss function, the loss will be zero when $|y_{i+q} - f(x_i)| \leq \epsilon$, and increase linearly at the rate C for points where the predicted value falls outside the ϵ-insensitive region. Based on statistical learning theory, this ϵ-SVR formulation explicitly provides robustness against parameter-driven model overfitting through the judicious choice of the regularization and bandwidth parameters (Smola & Schölkopf, 2004).[5] Appendix B further details the Lagrangian of the loss function, Eq. (2), and its estimation.

Note that in minimizing Eq. (2), the modeler is only free to adjust w, ξ_i, and ξ_i^*. The aim is to obtain w needed for constructing predictive models of the target variable, Eq. (1). Two basic tuning parameters, C and ϵ, together with any kernel-specific parameters in Υ, Eq. (1), serve here as inputs; see Table 1 for examples of kernel-specific parameter sets. Consequently, the SVR model forecasting performance depends upon the choice of parameters C and ϵ, along with any kernel-specific parameters in Υ. Using a kernel function with a set of tuning parameters, the SVR approach produces a single forecasting model. However, changing the tuning parameters would typically yield a different forecasting model.

2.2. Review of existing approaches to determining tuning parameters

Despite the high sensitivity of SVR OOS forecasting performance to the selection of kernel parameters, there is limited theoretical or generally accepted practical guidance on how to set these parameters. A number of approaches have been proposed in an attempt to address the issue.

Among various competing approaches, several authors set tuning parameters using a priori knowledge; see for example Vapnik (1995), Cherkassky and Mulier (2007), and Gunduz and Uhrig-Homburg (2011). However, a major downside of this approach is that its application is limited to expert users, with the outcome dependent on their skill level. Smola et al. (1998) and Kwok and Tsang (2003) propose an asymptotically optimal selection of the kernel bandwidth parameter ϵ values proportional to noise variance, an approach consistent with Vapnik (1995) and Cherkassky and Mulier (2007).[6] However, this approach assumes that the noise model is known to the modeler, which generally is not the case, limiting its practical application (Smola & Schölkopf, 2004). This approach also disregards the problem of the simultaneous estimation of the remaining kernel parameters (Chen & Wang, 2007). In contrast, several studies propose simple trial-and-error-based cross-validation (Bhatia, 2019; Cherkassky & Mulier, 2007; Schölkopf et al., 1999). However, this highly computationally intensive and time-consuming method, which also requires some luck finding satisfactory solutions, has been deemed impractical for many applications (Lins et al., 2012; Zhao et al., 2013).

In the financial time series literature, the alternative grid search algorithm has become the most frequently adopted technique; see for example Min and Lee (2005), Ding et al. (2008), Dai et al. (2013), and Dash et al. (2023). The grid search algorithm represents the parameter space as an evenly spaced collection of points within a certain range, with each point in the grid representing a choice for the tuning parameters. The optimal set of parameters is then obtained by searching through all the points in the grid. However, this technique, frequently combined with k-fold cross-validation, is sensitive to the size of the search step, time consuming, and computationally costly (Chen & Wang, 2007; Sun et al., 2021). It also requires a priori information about good initial parameter values, and similar to cross-validation, it does not guarantee finding a satisfactory solution (Lins et al., 2012).

To reduce the computational burden and improve the likelihood of finding a global solution, several authors have employed hybrid techniques that implement SVR in

[5] In an alternative ν-SVR formulation, the modeler specifies parameter ν to determine an upper bound on the fraction of observations which can fall outside the ϵ-insensitive region and a lower bound of the fraction of support vectors (Scholkopf et al., 2000). Then, ν is used to determine C and ϵ. Since there is no strong prior available regarding an appropriate value for ν in our application, we have applied ϵ-SVR.

[6] See Section 2.1 and Eqs. (1) and (2) for the description of the bandwidth parameter.

conjunction with a biologically inspired global optimization technique, with a genetic algorithm and krill herd optimization put forward by Holland (1992) and Gandomi and Alavi (2012), respectively, and subsequently adopted in financial market studies. Inspired by the theory of evolution, genetic algorithms simultaneously search for the global optimum parameter values in the nonlinear search space by mimicking natural selection and evolution processes. The algorithm encodes sets of potential parameter values using a chromosome-like data structure, using some recombination operator that imitates survival of the fittest, crossover, and mutation processes to find the optimal values (Whitley, 1994). Genetic algorithms have been adapted to forecasting Taiwan's stock market index in Chen and Ho (2005), exchange rates in Pai et al. (2006) and Sermpinis et al. (2015), and stock market volatility in Santamaría-Bonfil et al. (2015). However, these algorithms suffer from instability and are prone to converging to a local optimum, with their performance sensitive to the level of noise in the data, which can be high in financial data series (Santamaría-Bonfil et al., 2015; Sivaraj & Ravichandran, 2011; Zhao et al., 2013).

The alternative krill herd algorithm is based on the simulation of the collective feeding behavior of individual krill, with its objective function aiming to balance the stochastic random search between local and global optima, linked to distances of krill individuals from food and from the highest density of the swarm. Stasinakis et al. (2016) adapts krill herd optimization for forecasting commodity exchange-traded fund (ETF) returns and trading. Even though the evidence suggests that the krill herd algorithm performs better than other swarm and genetic algorithms, issues remain around its robustness and reliability at finding a global optimum, attributed to its poor exploitation and regulation between the exploration and exploitation processes (Li et al., 2014; Wang et al., 2014). To improve the balance between the global and local search, Sermpinis, Stasinakis, and Hassanniakalager (2017) propose a reverse adaptive krill herd locally weighted search method for forecasting European stock market ETFs, with alternative modifications put forward by Wang et al. (2013), Li et al. (2014), and Wang et al. (2014). In a distinct meta-heuristic application to predicting credit spreads of selected eurozone countries, Da Silva Fernandes et al. (2019) adapt the population-based sine cosine algorithm proposed by Mirjalili (2016) that employs a mathematical objective function rather than the biologically inspired one to improve the balance between the exploration and exploitation of the search space for finding the global optimum solution.

Bayesian optimization has recently been employed as another important technique for determining machine learning tuning parameters, found especially useful for large, complex systems that emerge, for example, when teams develop software systems for their clients and their corporate coworkers; see Shahriari et al. (2016) for a recent review of Bayesian optimization methods and their applications. The Bayesian approach develops a probabilistic surrogate model consisting of a prior distribution characterizing the beliefs about the behavior of the unknown objective function. The approach uses the information collected in the course of monitoring the objective function to update the priors in order to improve the likelihood of locating the best parameter settings. Among its few applications to predicting financial market price series, Law and Shawe-Taylor (2017) build upon Gao et al. (2002) in assuming the existence of a single most likely SVR model. But, as the direct application of Gao's approach delivers unsatisfactory results, the authors introduce a kernel scaling parameter in conjunction with the multi-armed bandit Bayesian optimization technique to improve and automate the parameter selection. However, the resulting mapping from the data to the tuning parameters is highly non-linear and extremely challenging to compute.[7] More generally, the performance of the Bayesian approach relies on the user's skill level and can suffer from a 'cold start' problem when the tuning application is new. Further, a bad choice of initial parameter values can make the optimization computationally costly.

Despite the improvements found in recently proposed approaches to setting tuning parameters, a common feature in all these techniques is that they aim to find one 'best' tuning parameter setting. We contribute to the literature by proposing an alternative approach that bypasses the challenge of finding the unique global optimum of the objective function and the single 'best' kernel parameter setting by accommodating a collection of good parameter settings associated with ubiquitous local optima. Our approach is similar in flavor and proposes an alternative to the Bayesian optimization approach that generates the prediction by averaging it over potential tuning parameter values, essentially integrating out these parameters. Our point of departure allows us to enhance the overall predictive performance by combining the forecasts produced by plausible candidates that likely contain complementary information about the data, achieving a consensus rather than selecting a single 'best'.

3. Forecasting framework

Our forecasting framework entails a two-step process. Step 1 involves specifying predictive models and setting SVR kernel function parameters, with the models subsequently estimated using the standard SVR methodology. The models' OOS forecasting accuracy is assessed in Step 2, enabling the selection of a subset of best-performing models. Using the data sample containing T observations, the first T_0 consecutive observations are allocated for training in Step 1, with $I = T_0$ in Eq. (1), and the remaining $T_1 = T - T_0$ observations reserved for OOS forecast evaluation in Step 2.

The forecasts are generated using rolling regressions. Depending on the modeler's objectives, the best-performing forecasts from Step 2 can subsequently be used directly or as inputs in forecast combinations. We proceed by outlining our methodology for setting the SVR tuning parameters and model selection and evaluation.

[7] To simplify the problem, the authors employ a Taylor-series approximation for the relationship, performing global quasi-optimization to find the most likely values of tuning parameters.

3.1. Setting tuning parameters

This section outlines the procedure we propose for selecting tuning parameter candidates, Υ, that serve as input into Eq. (1). Furthermore, as SVR forecasting algorithms routinely employ cross-validation procedures to ensure the robustness of forecasts, we outline our choice of cross-validation metrics, followed by a description of our procedure for selecting tuning parameters.

Cross-validation metrics

The presence of serial correlation in forecast errors of credit spreads, interest rates, and some other variables calls for selecting an appropriate cross-validation metric when evaluating the expected predictive performance of a given set of tuning parameters. This is because standard techniques can interpret serial correlation as a high-frequency relationship with small variance, which can result in setting ϵ too low for tracking the function variation (Brabanter et al., 2011). We employ the hv-block cross-validation metric proposed by Racine (2000), supported by evidence in the machine learning context in Bergmeir and Benítez (2012) and Cerqueira and Mozetič (2020).

The hv-block cross-validation approach provides a way to estimate the accuracy of forecasts in the presence of auto-correlated forecast errors. Consider a training data set $\mathcal{X} = \begin{bmatrix} x \\ y \end{bmatrix}$.[8] To apply hv-block cross-validation, for each observation i we collect v observations on either side of the observation $\mathcal{X}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}$ to construct a local validation set for time $t = i$ of size $n_v = 2v + 1$. Then, h observations are removed from either side of the local validation set, with the remaining $n_c = n - 2v - 2h - 1$ observations forming the local training set for time i.[9] The algorithm then trains on each local training set of size n_c, and computes errors on each local validation set of size n_v. Here, h aims to limit the impact of auto-correlation, maintaining near independence between the validation and training sets by h-blocking, whereas v ensures the consistency of the estimated forecast error variance.

Denoting the validation data subset by $\mathcal{X}_{(i:v)} = (y_{(i:v)}, x_{(i:v)})$, the hv-block cross-validation estimate of the forecast error variance is given by

$$\Phi(\Upsilon) = \frac{1}{(n-2v)n_v} \sum_{i=v+1}^{n-v} \|y_{(i:v)} - \widehat{y_{(i:v)}}(\Upsilon)\|^2. \tag{3}$$

Multi-sequential number-theoretic optimization

When searching for optimal tuning parameter values, the modeler faces an optimization problem that can have many local minima that are difficult to rank with any degree of certainty. To illustrate the problem, Fig. 1 plots $-\Phi(\Upsilon)$-surface, Eq. (3), for Mexico as an example, using the Baseline model with a linear kernel.[10] The figure shows many widely dispersed peaks representing local minima of the objective function, with no clear dominating, or 'best', point among the parameter values. Furthermore, we do not have a way to guess, with any precision, the regions where good parameter values might lie. Consequently, to mitigate the risk of neglecting favorable tuning parameters, we employ a robust global optimization technique, namely, multi-sequential number-theoretic optimization (MSNTO) proposed by Xu et al. (2005). Unlike many traditional numerical optimization methods designed to locate a single best solution, MSNTO can accommodate functions with many optima. It can search broadly and consequently is much less likely to be confounded by a bad initial search region.[11]

Consider a multivariate domain containing all of the parameter values we care to entertain $\pi = [a, b] \in \mathcal{R}^l$, $a_\iota \le \Upsilon_\iota \le b_\iota$, for all $\iota \in \{1, \ldots, l\}$ and a function, $\Phi(\Upsilon)$, continuous on π. We seek to find

$$\Upsilon^* \in \pi \text{ such that } \Upsilon^* = \Phi(\Upsilon^*) = \min_{\Upsilon \in \pi} \Phi(\Upsilon).$$

We choose a set $\Upsilon = \{\Upsilon_\iota, \iota = 1, \ldots, \Lambda\}$ widely dispersed on π, approximating Υ^* by $\widehat{\Upsilon^*_\Lambda} \in \Upsilon$ such that

$$\Phi(\widehat{\Upsilon^*_\Lambda}) = \min_{\iota \in \{1,\ldots,\Lambda\}} \Phi(\Upsilon_\iota).$$

This simple strategy is known to converge slowly. Recursively contracting the search region produces dramatic speed improvements. However, this sequential best-choice scheme might find only a local optimum. To avoid getting stuck at a local optimum, we employ the MSNTO algorithm that generates sample points using a number-theoretic sequence that maximizes dispersion. Then, at each step, the algorithm retains a small fraction, Π, of these points for clustering. Since the exact dispersion evaluation would be costly, the algorithm uses an approximation for dispersion:

$$\widehat{\mathcal{B}(\rho, \pi)} = \max_{1 \le \iota \le \Lambda} \min_{1 \le \theta \le \Lambda, \iota \ne \theta} \|\Upsilon_\iota - \Upsilon_\theta\|,$$

and calculates $\rho = \rho_\Phi \widehat{\mathcal{B}(\rho, \pi)}$, a small multiple of the estimate of dispersion. Here, ρ determines the range of points that should be included in a single cluster. The algorithm applies a user-supplied parameter σ to determine the maximum amount of contraction toward the local minimum to apply to each of the cluster regions.

These new cluster regions replace the old search regions during the next recursion. Xu et al. (2005) show that starting out using a number-theoretic sequence with a very large set of points, Λ_1, and continuing the recursion with a somewhat smaller set of points in the sequence, Λ_2, improves the algorithm's performance. We use $\Lambda_1 = 100$, $\Lambda_2 = 15$, $\Pi = 0.1$, $\rho_\Phi = 1.2$, $\sigma = 0.5$, the number of recursions $N = 3$, and generate points using a Niedrreiter sequence (Niederreiter, 1983). Fig. 2 further illustrates this process for a two-dimensional $\Phi(C, \epsilon)$

[8] Note that the cross-validation procedure only uses data from within the original training data set.

[9] The actual count remaining will increase when the observation is near the beginning or end of the available data, as one of the holdout sections shrinks.

[10] Section 4 and Table 2 therein provide details of the Baseline model of credit spreads.

[11] As the authors have done here, alternative robust optimization techniques can be similarly modified to retain potentially useful local optima.

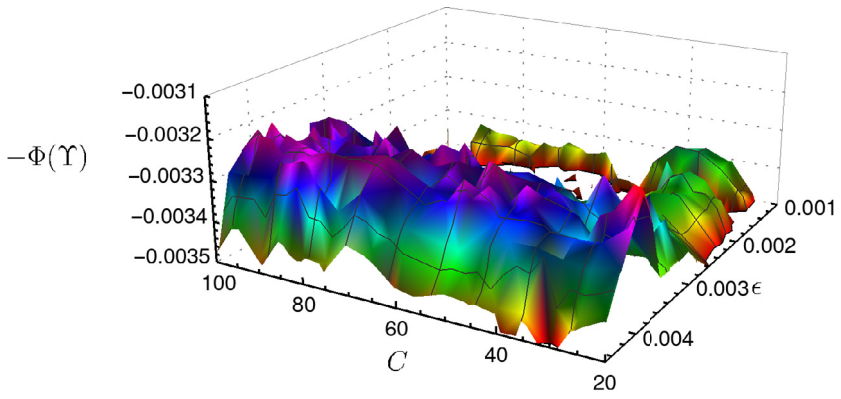


Fig. 1. $-\Phi(\Upsilon)$-surface, Eq. (3), for Mexico: Baseline model with a linear kernel.

Table 2
Predictor variables in the forecasting models.

Input variables	Models		
	Baseline	G	GEM
Country credit-spread-curve factors (level, slope, curvature)	✓	✓	✓
Global predictors			
US yield curve factors (level, slope, curvature)		✓	✓
Volatility of US short-term rate		✓	✓
Domestic predictors			
Financial risk rating			✓
Trade balance level			✓
Trade balance volatility			✓
Terms-of-trade growth level			✓
Terms-of-trade growth volatility			✓

parameter search using an SVR with a linear kernel as an example. The algorithm constructs clusters of points around each local minimum. The first set of points is widely dispersed, covering a large region of the search space. The second iteration has identified two clusters and retains the points associated with the two clusters. The third iteration further refines the search to clusters that encompass a smaller region, thereby increasing the precision of the approximation for the location of the two local minima. Accordingly, recursively applying the MSNTO algorithm produces clusters of tuning parameter sets in regions near local maxima, as can be observed in Fig. 2. In particular, when using a linear kernel function in combination with the Baseline model, illustrated in Fig. 1, the recursive MSNTO iterations produce clusters of (C, ϵ) pairs in regions near the many diverse local maxima. Consequently, as an output, the algorithm generates multiple (C, ϵ) pairs, each typically yielding a distinct forecasting model.

3.2. Model selection and evaluation

Our approach assesses the statistical significance of gains in predictive accuracy by means of the MCS test of Hansen et al. (2011). The procedure uses forecast errors to identify a subset of a group of models, called a model confidence set, constructed to contain all the superior models with a specified level of confidence. The MCS test has a number of important advantages over the widely used alternative tests put forward by Diebold and Mariano (1995), White (2000), and Hansen (2005). First, the MCS test provides additional information that is useful to the modeler: a measure of uncertainty surrounding model selection. The second advantage is related to the MCS test sensitivity to the utility of information in the available data, such that informative data produce a small collection of good models, whereas uninformative data generate large model confidence sets. Third, unlike alternative tests of pair-wise model comparisons, the MCS test does not require a benchmark, facilitating direct comparisons of forecasting accuracy among multiple competing model candidates, which is particularly useful in our case.

Our subsequent analysis employs the expected value of the squared error loss to evaluate the predictive ability of the OOS forecasts generated by various SVR candidate models.[12] Let the squared error loss function for the model j' prediction $\hat{y}_{j',t}$ of y_t be given by $L_{j',t} = L(y_t, \hat{y}_{j',t}) = (\hat{y}_{j',t} - y_t)^2$. Define the measure of relative model performance as $\mu_{j'j''} \equiv E(L_{j',t} - L_{j'',t})$. Thus, model j' is preferred over model j'' when $\mu_{j'j''} < 0$. The authors assume that $\mu_{j'j''}$ is finite and independent of t.

To define the MCS test procedure, consider a finite initial collection of forecasting models, $\mathcal{M}^0$. Let $\mathcal{M}^*$ denote the set of best models for a specified metric of model assessment:

$$\mathcal{M}^* \equiv \{j' \in \mathcal{M}^0 : \mu_{j'j''} \leq 0 \ \text{ for all } j'' \in \mathcal{M}^0\}.$$

A model confidence set at level α, $\mathcal{M}^*_{1-\alpha}$, is then a subset of $\mathcal{M}^0$ containing all of $\mathcal{M}^*$ with a probability $(1-\alpha)$. For a given loss function and confidence level, the test uses sample information about each model's relative performance OOS to sequentially eliminate the poorest-performing models, producing p-values for each model in $\mathcal{M}^0$. The test procedure estimates $\widehat{\mathcal{M}}^*_{1-\alpha}$ via a sequence of significance tests with the null hypothesis $H_{0,\mathcal{M}} : \mu_{j'j''} = 0$ for all $j', j'' \in \mathcal{M}$; $\mathcal{M} \subset \mathcal{M}^0$ and alternative $H_{A,\mathcal{M}} : \mu_{j'j''} \neq 0$ for some $j', j'' \in \mathcal{M}$. The sequential elimination

[12] See Hansen et al. (2011) for alternative loss functions that the MCS test accommodates.

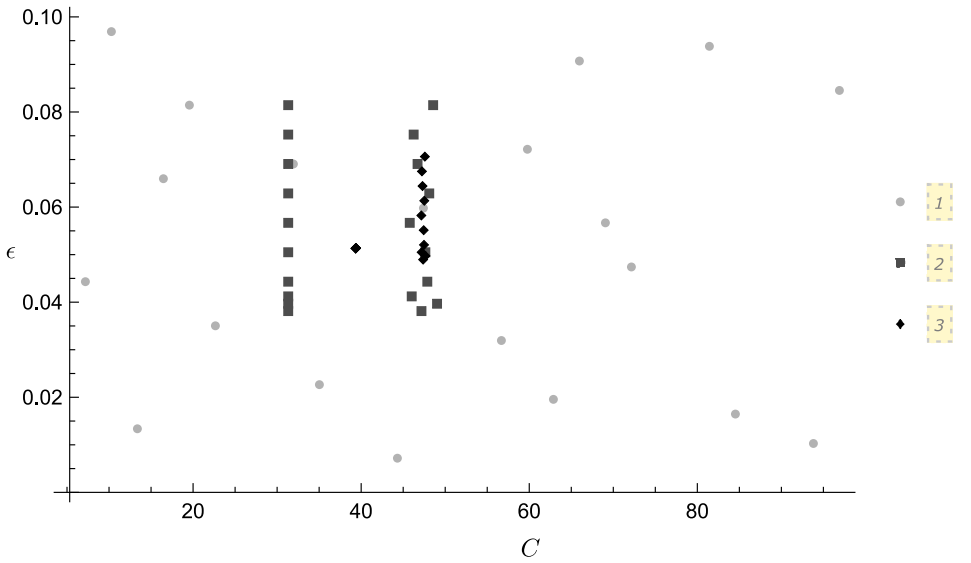


Fig. 2. MSNTO ($-\Phi(C,\epsilon)$) evaluation point clustering and contraction: Example from an SVR with a linear kernel.

from the model confidence set continues until doing so reduces the coverage ratio, $(1-\alpha)$, below the specified confidence level.[13]

The authors show that, since the procedure uses the same significance level in all tests, all models with p-values greater than α are in $\widehat{\mathcal{M}}^{*}_{1-\alpha}$. When the test assigns relatively high p-values to only one or a few models, this serves as compelling evidence of their superior predictive accuracy relative to the competitor models. We employ the Hansen et al. (2011) maximum t-statistic test for assessing model forecasting accuracy using OOS forecast errors from rolling regression estimation.[14]

Forecast combinations

While a collection of competitive forecasts generated by our approach can be of ultimate interest to some users, many applications require a single 'best' forecast. Such competitive forecasts can be obtained by combining forecasts generated by our best model sets. Recent evidence in forecast combination studies suggests that combining forecasts can deliver gains in predictive accuracy (Paye, 2012; Rapach et al., 2010). Timmermann (2006) and Rapach et al. (2010), inter alia, further show that simple forecast combining methods such as the mean forecast combination perform well in some financial applications. Furthermore, combining forecasts from our best model sets, selected from a collection of candidates by means of the MCS test as detailed in Section 3.2, is consistent with the MCS test-based model trimming approach in Samuels and Sekkel (2017) who provide evidence of its competitive forecasting performance relative to other trimming methods.

Consequently, we follow Rapach et al. (2010) and Paye (2012) and construct two simple forecast combinations, namely, the mean forecast combination and the median forecast combination, for evaluating the predictive accuracy of the high-quality model sets generated by our approach.[15] Accordingly, the combination forecast is the weighted average of the N individual forecasts generated by the set of MSNTO-based SVR predictive models with superior predictive accuracy. The mean forecast combination employs equal weights of $\frac{1}{N}$, whereas the median forecast combination is the median of $\{y^{i}_{c,t+q}\}_{i=1}^{N}$, with i referring to a given model from the best model set.

We identify the best model sets for constructing our forecast combinations using rolling samples of individual OOS candidate MSNTO-based SVR forecasts, whereby the best model set is successively updated for each subsequent forecast. More specifically, the first $T_{1,0}$ forecasts from our OOS forecast evaluation sample of T_1 observations are utilized for selecting the set of best models that enter the model combination, producing our first combination forecast. Next, we roll our sample forward by one observation and repeat the model selection step, enabling our second combination forecast, and so forth. This procedure is iterated forward to produce a sample of

[13] The procedure uses bootstrapping methods to implement the testing procedure. Even though Hansen et al. (2011) primarily focus on parametric models, numerous studies have shown that SVRs using standard kernels such as linear functions, RBFs, and polynomial functions have good statistical and numerical properties, such as universal consistency under weak assumptions, as well as good generalization properties in many applications, pointing to the validity of these bootstrap methods for the SVRs; see Vapnik (1995), Vapnik (1998), Scholkopf and Smola (2001), Steinwart and Christmann (2008), Hable (2012), and Christmann and Hable (2013). Furthermore, computing standard measures for monitoring bootstrap behavior in our application revealed that the SVR bootstraps are generally well behaved and comparable to ordinary least squares (OLS) bootstraps (Efron & Tibshirani, 1993; Hansen, 2011; Timmermann & Zhu, 2019).

[14] We used the *ForecastEval* package, a Julia implementation of the MCS test.

[15] Wang et al. (2023) provide a recent review of alternative forecast combining methods. We leave the testing and application of more sophisticated methods for future research.

$T_{1,1}$ ($T_1 - T_{1,0} - q$) OOS MSNTO-based SVR combination forecasts.

To assess the sensitivity of forecast accuracy to the statistical level of confidence utilized in the selection of the MSNTO-based SVR model sets entering forecast combinations, we consider a range of combination forecasts based on the best $\widehat{\mathcal{M}}^*_{10\%}$, $\widehat{\mathcal{M}}^*_{20\%}$, $\widehat{\mathcal{M}}^*_{40\%}$, $\widehat{\mathcal{M}}^*_{60\%}$, $\widehat{\mathcal{M}}^*_{80\%}$, and $\widehat{\mathcal{M}}^*_{90\%}$ model sets. We abbreviate these as MSNTO-Mean-$\widehat{\mathcal{M}}^*_{10\%}$, MSNTO-Median-$\widehat{\mathcal{M}}^*_{10\%}$, MSNTO-Mean-$\widehat{\mathcal{M}}^*_{20\%}$, MSNTO-Median-$\widehat{\mathcal{M}}^*_{20\%}$, and so forth.

Forecast benchmarking

To contrast the accuracy of our MSNTO forecasts with that of standard SVR specifications, we employ a variety of SVR benchmarks, adding them to the list of competing forecasts in our MCS test-based forecast evaluations. For completeness, we further augment our benchmark set by random forests model forecasts, along with several established benchmarks for the empirical application at hand.

4. Empirical application: Forecasting EME credit spreads

In our empirical application, we employ the weekly data set of Audzeyeva and Fuertes (2018). Below, we briefly introduce our target variable and predictor variables, followed by the details of the empirical application of our approach and the description of benchmarks.[16]

4.1. Emerging market credit spreads and spread curve factors

The target variable in our predictability analysis, Eq. (1), is the credit spread on sovereign bonds of an emerging market country c:

$$y_{c,t}(\tau) \equiv Y_{c,t}(\tau) - Y_{US,t}(\tau), \tag{4}$$

where $Y_{c,t}(\tau)$ and $Y_{US,t}(\tau)$ are the time-t yield to maturity on τ-maturity zero-coupon bonds of emerging-market country c and the U.S. Treasury, respectively. In our predictive analysis, we set $\tau = 5$ years, at which EME bonds are highly liquid.[17] The data set contains the yields to maturity on zero-coupon U.S. Treasuries and U.S. dollar-denominated euro-bonds of each country we used to calculate the country-specific credit spreads.

Adopting the Nelson and Siegel (1987) representation permits expressing the credit spread on τ-maturity zero-coupon bond of sovereign c as a parsimonious function of its credit-spread-curve factors:

$$y_{c,t}(\tau) = \beta_{c0,t} + \beta_{c1,t}\left(\frac{1 - e^{-\lambda_{c,t}\tau}}{\lambda_{c,t}\tau}\right) + \beta_{c2,t}\left(\frac{1 - e^{-\lambda_{c,t}\tau}}{\lambda_{c,t}\tau} - e^{-\lambda_{c,t}\tau}\right) \tag{5}$$

where $\beta_{c0,t}$, $\beta_{c1,t}$, and $\beta_{c2,t}$, are the level, slope, and curvature factors, respectively. The country credit-spread-curve factors are available at a weekly frequency in our data set.[18]

Fig. 3 plots the evolution of credit spread curves for Brazil, Mexico, the Philippines, and Turkey during our December 2, 2008, to December 29, 2015 sample period while Table A.1 in Appendix A gives summary statistics for the target variable, five-year credit spreads, and the credit-spread-curve factors. Credit spreads and credit-spread-curve factors of all countries exhibit stylized persistence. Fig. 3 further shows that although the dynamics of credit spreads exhibit common trends across the four emerging market economies, driven by various global market factors, country-specific variations are nevertheless apparent. Such variations primarily reflect differences in country-specific credit risk. In particular, in December 2008, Brazil and Mexico (both investment-grade BBB-rated by the S&P credit rating agency) enjoyed lower credit spreads than the Philippines and Turkey, rated as speculative-grade BB. When Mexico's and the Philippines' credit ratings remained stable, and their spreads experienced a long-term downward trend, Brazil's spreads rose sharply with the S&P issuing a negative rating outlook for Brazil on June 6, 2013, followed by several negative rating changes that led to Brazil's downgrade to speculative-grade rating BB on September 9, 2015. Turkey also experienced jumps in its credit spreads, following negative rating outlooks by the S&P between April 3, 2008, and February 7, 2014, despite its credit rating remaining unchanged. All three speculative-grade borrowers exhibit higher mean credit spreads and volatility relative to investment-grade Mexico.

4.2. Predictor variables

Variable selection for our predictive models is rooted in financial theory and empirical evidence from the bond market OOS forecasting literature. More specifically, our Baseline model employs as input variables the country-specific level, slope, and curvature factors $(\beta_{c0,t}, \beta_{c1,t}, \beta_{c2,t})$ that jointly summarize the information content in today's credit spread curve. The Baseline model is rooted in the rational expectations theory of the term structure of interest rates of Sargent (1973) and Roll (1970). The main idea is that investor expectations about future credit spreads, reflected in today's forward spreads, exploit all the available information. Accordingly, today's credit spread curve, which embeds forward credit spreads, contains relevant information for predicting future spreads. As it reflects investor expectations about the future path

[16] Audzeyeva and Fuertes (2018) provide detailed descriptions of the variable selection, data, and data sources.

[17] Bonds at the low end and high end of the maturity spectrum tend to be traded less actively, which can affect their pricing.

[18] Audzeyeva and Fuertes (2018) extract the credit curve level, slope, and curvature factors from weekly cross-sections of country-specific bond prices; see their paper for details.

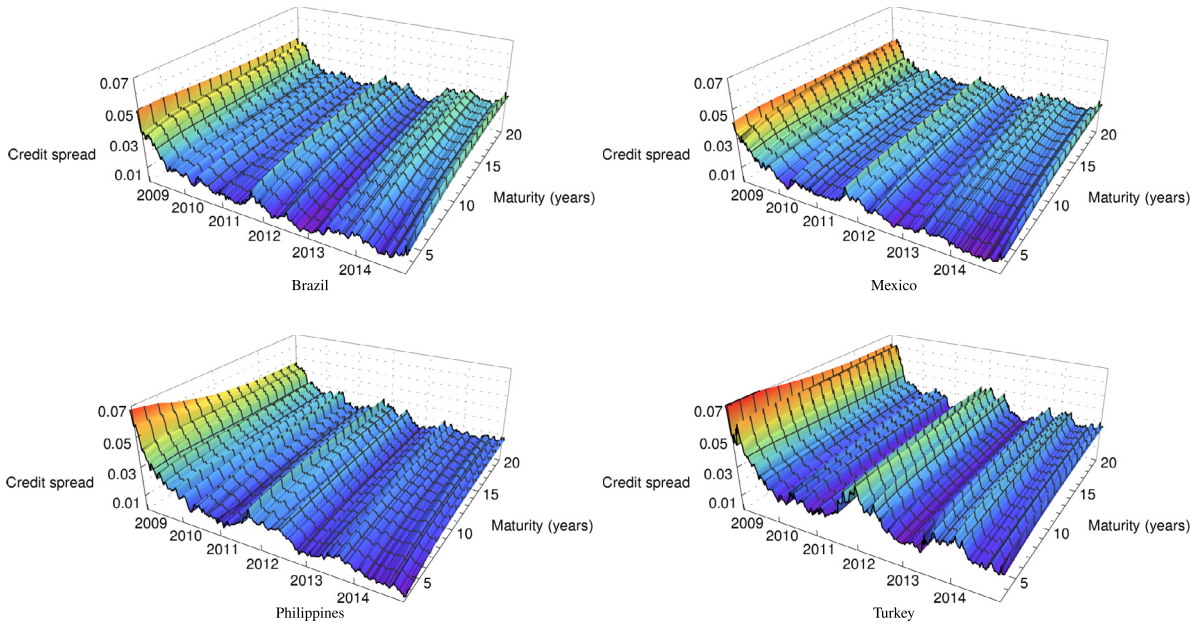


Fig. 3. Emerging-market credit spreads.

of credit spreads, it serves as a forward-looking predictor. The supportive evidence of significant predictive content in the cross-section of current bond yields and the superior predictive OOS performance of the parsimonious Nelson and Siegel (1987) representation relative to alternative specifications is well documented in the context of developed-economy bond markets in Cochrane and Piazzesi (2005), Diebold and Li (2006), Diebold et al. (2008), and Christensen et al. (2011), among others.

We further employ two alternative model specifications that augment the input vector in the Baseline model by global macroeconomic variables, model G, and both global and domestic macroeconomic variables, model GEM. Table 2 lists input variables in each predictive model and Table A.1 in Appendix A provides summary statistics of the data variables.

Our first global predictor is the global riskless yield curve, summarized by the level, slope, and curvature factors of the U.S. Treasury yield curve.[19] Global interest rates are known to strongly influence emerging market credit spreads via multiple channels, for example by impacting the sovereign borrower's domestic economic conditions, thereby affecting the sovereign's future ability to repay external debt, and by influencing international investors' risk appetite that, in turn, determines their demand for emerging-market bonds (e.g. Ciarlone et al., 2009; Gibson & Sundaresan, 2005; Hartelius et al., 2008; Uribe & Yue, 2006). The global riskless yield curve captures forward-looking information content, reflecting investor expectations about the future path of global interest rates, with their significant predictive ability documented for both developed-economy sovereign bonds and corporate bonds by Diebold et al. (2008) and Krishnan et al. (2010), respectively. Krishnan et al. (2010) provides further evidence that the entity-specific credit spread curve and global riskless yield curve jointly impound all relevant information for predicting future credit spreads. The U.S. short-term interest rate volatility is our second global predictor, which captures investor uncertainty about global economic conditions, thereby affecting global market liquidity and financial risk allocation by international investors, both relevant for credit spreads (Arora & Cerisola, 2001; Hartelius et al., 2008).[20]

Motivated by findings in the macro-finance literature that domestic macroeconomic variables can improve predictions based on the bond yield curve in the context of developed economies, we expand our variable set by a set of domestic predictors adapted to our EME context (Ang & Piazzesi, 2003; Huang et al., 2023; Ludvigson & Ng, 2009; Moench, 2008). Our first domestic variable is the country financial risk rating, constructed by the International Country Risk Guide of the Political Risk Services Group (PRS) using various sovereign-debt-related indicators that signal the more imminent risk of sovereign default affecting sovereign credit spreads (e.g. Eichengreen & Mody, 1998; Min, 1998).[21] Audzeyeva and Fuertes (2018) find that that PRS rating changes lead by roughly four months the rating change announcements by international rating agencies such as S&P, making it our preferred predictor. The next four domestic variables are informed by business-cycle theory and empirical evidence that emphasizes the importance of the external sector of

[19] Similar to the country credit-spread-curve factors, weekly U.S. Treasury yield-curve factors are extracted from weekly cross-sections of U.S. Treasury bond prices.

[20] The U.S. short rate volatility is measured by the standard deviation of the one-year U.S. Treasury yield over the preceding 10-week window.

[21] Csonto and Ivaschenko (2013) provide a detailed description of the PRS country financial risk ratings.

an emerging market economy in signaling its future ability to repay external debt (Aguiar & Gopinath, 2006; Chen & Rogoff, 2003; Hilscher & Nosbusch, 2010; Mendoza, 1995; Neumeyer & Perri, 2005). Accordingly, the relevant external sector variables are trade balance, trade balance volatility, terms-of-trade growth, and terms-of-trade growth volatility.[22,23]

4.3. Empirical application of our forecasting approach

For the OOS predictability analysis of EME credit spreads, we allocate for training, in Step 1 of our approach outlined in Section 3, the first $T_0 = 2/3T$ consecutive weeks of the $T = 367$ weekly observations available in the data sample period, setting aside the remaining $T_1 = 1/3T$ weeks for OOS forecast evaluation in Step 2 (Hansen et al., 2011). We set a predictive horizon of $q = 13$ weeks, or one quarter, in Eq. (1), as in Audzeyeva and Fuertes (2018). A quarter-ahead predictive horizon is aligned with our focus on longer-term investors and the nature of term structure models using macroeconomic data variables that we employ.

When implementing our parameter tuning procedure in Step 1 using our training data set of 244 observations, we set $h = 2$, $v = 7$ in Eq. (3), informed by Racine (2000) analysis for medium-sized data sets.

To construct forecast combinations using forecasts generated by our best model sets from Step 2, we identify the best model sets for constructing our forecast combinations using rolling samples of individual OOS candidate MSNTO-based SVR forecasts. More specifically, the first $T_{1,0} = 52$ weekly forecasts from our OOS forecast evaluation sample of $T_1 = 127$ weeks is utilized for selecting the set of best models that enter the model combination, producing our first quarter-ahead ($q = 13$ weeks) combination forecast. Next, we roll our sample one week forward and repeat the model selection step, enabling our second quarter-ahead combination forecast, and so forth. This procedure is iterated forward to produce a sample of $T_{1,1} = 62$ ($T_1 - T_{1,0} - q$) OOS MSNTO-based SVR combination forecasts. Furthermore, Appendix C provides pseudocode aiding a step-by-step practical implementation of our forecasting approach.

4.4. Forecast benchmarking

Our benchmarking exercise employs several standard SVR benchmarks and random forests model forecasts, together with several established benchmarks from the credit spread literature that are outlined next.

Standard SVR models

We entertain a battery of standard SVR specifications that attempt to find the 'best' model for each kernel and set of input variables. To this end, we employ the popular grid search (GS) methodology for selecting the optimal tuning parameter set. We use the same search regions as those we used in the SVR kernel calculations. However, in contrast to the recursive clustering and search region refinement employed in our MSNTO calculations, we evaluate the cross-validation criteria at a comparable number of uniformly arrayed GS points, and choose the best point.

Accordingly, employing the same sets of predictors as for our main SVR forecasts detailed in Section 4.2, we estimate three benchmark SVR models for each country and kernel combination, producing altogether nine SVR benchmarks. We abbreviate them as GS-Linear-Baseline, GS-Linear-G, GS-Linear-GEM, GS-RBF-Baseline, GS-RBF-G, GS-RBF-GEM, GS-Poly-Baseline, GS-Poly-G, and GS-Poly-GEM.

Random forests model

Random forests (RF) models provide a relatively new tool for constructing robust and accurate predictors (Liu et al., 2012). As an ensemble learning implementation of decision trees, they serve to mitigate the overfitting problem often associated with that technique (Bramer, 2013). One can interpret RF as a convenient method for constructing and combining the forecasts of a complementary collection of individual decision trees. We used the default values supplied by the Mathematica implementation of RF regression for constructing the RF forecasts. This benchmark forecast can be regarded as a random forecast made by the 'naive' user based on the current value of the credit spread.

Random walk model

The random walk (RW) model represents the canonical benchmark originating from the sovereign bonds forecasting literature. An RW reflects the observation by Christensen et al. (2011) and Duffee (2011), among others, that the evolution of bond yields resembles a random walk. Duffee (2002) provides empirical evidence that an RW outperforms OOS forecasts from the standard affine equilibrium term structure models for developed-economy bonds, whereas Audzeyeva and Fuertes (2018) further documents that its forecasts are similarly formidable against various linear model forecasts in the context of EME bonds. In the RW model, the forecast is for no-change:

$$y_{c,t+q} = y_{c,t} + \nu_{c,t+q}.$$

Linear OLS-baseline, OLS-G, and OLS-GEM models

To represent another set of benchmarks, commonly utilized linear regression models, we employ three OLS regression models from Audzeyeva and Fuertes (2018) that use the same sets of input variables as our main SVR forecasts detailed in Section 4.2. Thus, we consider three additional OLS benchmark models.

[22] Trade balance is defined as exports minus imports over GDP in US$ in a given month, and terms-of-trade growth is defined as the annual percentage change in the US$ price of the country's exports relative to the US$ price of its imports. Volatility is measured by the standard deviation over the most recent six-month window. Weekly series are obtained by interpolation from monthly data.

[23] Data availability limitations for the Philippines precluded us from employing the terms-of-trade data in GEM for this borrower.

The first model, OLS-Baseline, generates forecasts that utilize the predictive content in the credit-spread-curve level, slope, and curvature factors:

$$y_{c,t+q} = \psi_c + \kappa_{c0}\hat{\beta}_{c0,t} + \kappa_{c1}\hat{\beta}_{c1,t} + \kappa_{c2}\hat{\beta}_{c2,t} + \nu_{c,t+q} \quad (6)$$

The second model, OLS-G, augments OLS-Baseline with the vector of global macroeconomic input variables, $\boldsymbol{G}_t$:

$$y_{c,t+q} = \psi_c + \kappa_{c0}\hat{\beta}_{c0,t} + \kappa_{c1}\hat{\beta}_{c0,t} + \kappa_{c2}\hat{\beta}_{c2,t} + \boldsymbol{\theta}_c^G\boldsymbol{G}_t + \nu_{c,t+q}. \quad (7)$$

In the third model, OLS-GEM, the input set is further augmented with the vector of domestic macroeconomic variables, $\boldsymbol{EM}_{c,t}$:

$$y_{c,t+q} = \psi_c + \kappa_{c0}\hat{\beta}_{c0,t} + \kappa_{c1}\hat{\beta}_{c0,t} + \kappa_{c2}\hat{\beta}_{c2,t} + \boldsymbol{\theta}_c^G\boldsymbol{G}_t + \boldsymbol{\theta}_c^{EM}\boldsymbol{EM}_{c,t} + \nu_{c,t+q}. \quad (8)$$

The predictive horizon is $q = 13$ weeks, as above. The benchmark predictions are obtained using rolling regressions based on the same training and OOS evaluation windows as our MSNTO-based SVR model predictions.

5. Empirical forecasting results

5.1. Predictive ability of MSNTO-based SVR forecasts

Using our OOS forecast evaluation sample (T_1), our first forecasting exercise identifies most accurate sets of MSNTO-based SVR predictive models by evaluating the accuracy of the quarter-ahead OOS forecasts generated by our collection of model candidates.[24] To contrast the accuracy of our MSNTO-based SVR forecasts with that of the benchmarks, we add the benchmark forecasts to our MCS tests. Table 3 reports the results for the 25 MSNTO-based SVR forecasts that have the lowest Root Mean Squared Error (RMSE) for each country, along with forecasts from the standard GS-based SVR, RF, RW, and OLS benchmarks.[25] Also, Table D.1 in Appendix D reports examples of tuning parameter values for the top five MSNTO-based SVR models in Table 3.

To gauge the statistical significance of gains in forecast accuracy, we report MCS p-values, focusing our subsequent discussion on model forecasts in $\widehat{\mathcal{M}}^*_{20\%}$, $\widehat{\mathcal{M}}^*_{40\%}$, and $\widehat{\mathcal{M}}^*_{65\%}$.[26] The interpretation of the MCS test confidence level is analogous to that of a confidence interval for a parameter where the MCS identifies, from a collection of model candidates, a subset of models that contains the best model with a given level of confidence. Model forecasts are ordered by MCS p-value, with those most likely to generate accurate forecasts listed first.

The wide range of reported p-values across models provides evidence of useful information content in each country's data that unambiguously identifies a subset of most accurate forecasts. More specifically, the OOS forecasting evidence clearly identifies relatively small sets of the MSNTO-based SVR models with superior predictive accuracy over the remaining candidates and the benchmarks. For example, six MSNTO-based forecasts but none of the benchmark forecasts enter $\widehat{\mathcal{M}}^*_{40\%}$ for Brazil. The evidence is even more striking for Mexico and the Philippines where, respectively, 14 and 15 MSNTO-based forecasts enter $\widehat{\mathcal{M}}^*_{20\%}$. At the same time, all benchmark forecasts exhibit relatively low p-values for the three countries, clearly signaling their inferior predictive accuracy when compared to our SVR forecasts. The only exception is the GS-RBF-GEM SVR benchmark forecast for Turkey that enters $\widehat{\mathcal{M}}^*_{20\%}$. Nevertheless, in line with evidence obtained for other countries, the remaining models in $\widehat{\mathcal{M}}^*_{20\%}$ as well as those in $\widehat{\mathcal{M}}^*_{40\%}$ and $\widehat{\mathcal{M}}^*_{65\%}$ for Turkey are all MSNTO-based SVR, with no further RF, SVR, OLS, or RW benchmarks making it into the superior model confidence sets.

This evidence highlights the limitations of standard SVR techniques, relying on finding one 'best' set of tuning parameters that leads to one 'best' model, lending support to our approach that delivers a model set with superior predictive accuracy by exploiting a *collection* of plausible parameter settings.

5.2. Predictive ability of MSNTO-based SVR forecast combinations

Next, we turn to evaluating the accuracy of the OOS forecast combinations generated by our best model sets by benchmarking them against a variety of the standard SVR, RF, RW, and OLS model forecasts. To this end, we employ the MCS test for the second time. The best model sets entering into the forecast combinations are updated weekly, in real time, as detailed in Section 3.2. Table 4 reports the results for the mean and median forecast combinations, based on the best model sets at various levels of confidence, and the benchmarks.

The *MCS* test results provide clear evidence that our MSNTO-based SVR forecast combinations deliver superior forecast accuracy, consistently outperforming the benchmarks across countries. In particular, only the mean and median MSNTO-based SVR combination forecasts but none of the benchmarks enter $\widehat{\mathcal{M}}^*_{20\%}$ for all countries. $\widehat{\mathcal{M}}^*_{40\%}$ and $\widehat{\mathcal{M}}^*_{65\%}$ are similarly dominated by the MSNTO combination forecasts, with none of the benchmarks making it into the most accurate model sets for Mexico and Turkey, and only a single country-specific benchmark, namely, OLS-GEM and RW, making it into $\widehat{\mathcal{M}}^*_{40\%}$ and $\widehat{\mathcal{M}}^*_{65\%}$, respectively, for Brazil and the Philippines. However, neither of these benchmarks deliver competitive predictive performance for other countries.

Gains in forecast accuracy are equally pronounced in economic terms, as borne out by a notable reduction in forecast errors delivered by our MSNTO-based SVR forecast combinations for all four countries. More specifically, the forecast combinations in $\widehat{\mathcal{M}}^*_{20\%}$ deliver, on average, a

[24] We investigate nine kernel-model combinations that underpin our forecasts, resulting from pairing three SVR kernels (Linear, RBF, and Polynomial) with three models (Baseline, G, and GEM).

[25] Our analysis (omitted here for brevity) has shown that entering larger sets of MSNTO-based SVR models into the MCS test did not materially change the most accurate model selection results and model MCS p-values reported in Table 3.

[26] The levels of confidence that we focus on are generally more restrictive than the 75% and 90% reported in the forecasting exercise by Hansen et al. (2011).

Table 3
Predictive ability of MSNTO-based SVR candidate models generated by our approach[a,b].

Brazil			Mexico			Philippines			Turkey		
Model	RMSE	P_{MCS}	Model	RMSE	P_{MCS}	Model	RMSE	P_{MCS}	Model	RMSE	P_{MCS}
MSNTO-RBF-GEM-1	0.460	1.000	MSNTO-Poly-Baseline-1	0.184	1.000	MSNTO-Poly-Baseline-1	0.181	1.000	MSNTO-RBF-GEM-1	0.330	1.000
MSNTO-RBF-GEM-2	0.460	0.748	MSNTO-Poly-Baseline-2	0.185	0.958	MSNTO-Poly-Baseline-2	0.182	0.870	GS-RBF-GEM	0.331	0.857
MSNTO-RBF-GEM-3	0.460	0.748	MSNTO-Poly-Baseline-3	0.186	0.958	MSNTO-Poly-Baseline-3	0.184	0.870	MSNTO-RBF-GEM-2	0.331	0.622
MSNTO-RBF-GEM-4	0.460	0.666	MSNTO-Poly-Baseline-4	0.189	0.912	MSNTO-Poly-Baseline-7	0.186	0.870	MSNTO-RBF-GEM-3	0.333	0.441
MSNTO-RBF-GEM-6	0.461	0.640	MSNTO-Poly-Baseline-5	0.189	0.908	MSNTO-Poly-Baseline-5	0.186	0.870	MSNTO-RBF-GEM-4	0.333	0.441
MSNTO-RBF-GEM-5	0.461	0.640	MSNTO-Poly-Baseline-9	0.194	0.908	MSNTO-Poly-Baseline-4	0.185	0.870	MSNTO-RBF-GEM-5	0.333	0.215
MSNTO-RBF-GEM-8	0.461	0.403	MSNTO-Poly-Baseline-6	0.191	0.846	MSNTO-Poly-Baseline-6	0.186	0.870	MSNTO-RBF-GEM-12	0.334	0.119
MSNTO-RBF-GEM-7	0.461	0.403	MSNTO-Poly-Baseline-8	0.193	0.846	MSNTO-Poly-Baseline-9	0.187	0.870	MSNTO-RBF-GEM-8	0.334	0.119
MSNTO-RBF-GEM-14	0.462	0.403	MSNTO-Poly-Baseline-11	0.194	0.846	MSNTO-Poly-Baseline-8	0.186	0.870	MSNTO-RBF-GEM-14	0.335	0.012
MSNTO-RBF-GEM-9	0.461	0.403	MSNTO-Poly-Baseline-10	0.194	0.840	MSNTO-Linear-Baseline-11	0.187	0.870	MSNTO-RBF-GEM-6	0.333	0.012
MSNTO-RBF-GEM-19	0.462	0.403	MSNTO-Linear-Baseline-7	0.193	0.840	MSNTO-Linear-Baseline-21	0.188	0.870	MSNTO-RBF-GEM-7	0.334	0.012
MSNTO-RBF-GEM-12	0.462	0.403	MSNTO-Poly-Baseline-12	0.195	0.840	MSNTO-Linear-Baseline-18	0.188	0.870	MSNTO-RBF-GEM-17	0.337	0.002
MSNTO-RBF-GEM-23	0.463	0.403	MSNTO-Poly-Baseline-16	0.196	0.840	MSNTO-Linear-Baseline-17	0.188	0.870	MSNTO-RBF-GEM-13	0.334	0.002
MSNTO-RBF-GEM-10	0.462	0.273	MSNTO-Poly-Baseline-18	0.196	0.840	MSNTO-Linear-Baseline-10	0.187	0.870	MSNTO-RBF-GEM-9	0.334	0.002
MSNTO-RBF-GEM-13	0.462	0.273	MSNTO-Poly-Baseline-17	0.196	0.778	MSNTO-Linear-Baseline-12	0.188	0.806	MSNTO-RBF-GEM-10	0.334	0.002
MSNTO-RBF-GEM-11	0.462	0.273	MSNTO-Poly-Baseline-14	0.195	0.778	MSNTO-Poly-Baseline-24	0.188	0.446	MSNTO-RBF-GEM-19	0.337	0.002
MSNTO-RBF-GEM-24	0.463	0.273	MSNTO-Poly-Baseline-20	0.198	0.778	MSNTO-Linear-Baseline-15	0.188	0.446	MSNTO-RBF-GEM-11	0.334	0.002
MSNTO-RBF-GEM-17	0.462	0.273	MSNTO-Poly-Baseline-24	0.200	0.778	MSNTO-Linear-Baseline-25	0.189	0.446	MSNTO-RBF-GEM-20	0.337	0.002
MSNTO-RBF-GEM-16	0.462	0.243	MSNTO-Poly-Baseline-15	0.196	0.658	MSNTO-Linear-Baseline-22	0.188	0.446	MSNTO-RBF-GEM-15	0.336	0.002
MSNTO-RBF-GEM-18	0.462	0.214	MSNTO-Poly-Baseline-13	0.195	0.658	MSNTO-Linear-Baseline-16	0.188	0.446	MSNTO-RBF-GEM-16	0.336	0.002
MSNTO-RBF-GEM-25	0.463	0.214	MSNTO-Poly-Baseline-21	0.198	0.658	MSNTO-Poly-Baseline-13	0.188	0.446	MSNTO-RBF-GEM-25	0.338	0.002
MSNTO-RBF-GEM-21	0.463	0.214	MSNTO-Poly-Baseline-19	0.196	0.658	MSNTO-Linear-Baseline-14	0.188	0.446	MSNTO-RBF-GEM-18	0.337	0.002
MSNTO-RBF-GEM-15	0.462	0.214	MSNTO-Linear-Baseline-23	0.199	0.658	MSNTO-Linear-Baseline-20	0.188	0.446	MSNTO-RBF-GEM-24	0.337	0.002
MSNTO-RBF-GEM-20	0.462	0.050	MSNTO-Poly-Baseline-22	0.199	0.200	MSNTO-Linear-Baseline-23	0.188	0.446	MSNTO-RBF-GEM-23	0.337	0.000
MSNTO-RBF-GEM-22	0.463	0.050	RF	0.214	0.200	MSNTO-Linear-Baseline-19	0.188	0.446	MSNTO-RBF-GEM-22	0.337	0.000
OLS-GEM	0.496	0.050	MSNTO-Poly-Baseline-25	0.200	0.197	RW	0.204	0.395	MSNTO-RBF-GEM-21	0.337	0.000
GS-RBF-G	0.523	0.050	OLS-GEM	0.215	0.197	GS-Poly-Baseline	0.194	0.290	GS-Linear-Baseline	0.416	0.000
OLS-G	0.520	0.050	GS-RBF-GEM	0.255	0.197	GS-RBF-GEM	0.238	0.290	OLS-Baseline	0.416	0.000
GS-Poly-GEM	0.528	0.050	OLS-G	0.227	0.197	GS-Linear-G	0.239	0.290	GS-RBF-G	0.449	0.000
OLS-Baseline	0.546	0.050	GS-RBF-Baseline	0.249	0.117	OLS-GEM	0.267	0.290	GS-Poly-Baseline	0.472	0.000
GS-Linear-GEM	0.535	0.050	RW	0.225	0.015	GS-Poly-G	0.267	0.290	GS-RBF-Baseline	0.438	0.000
GS-RBF-GEM	0.742	0.050	GS-Linear-Baseline	0.254	0.015	OLS-G	0.265	0.290	RW	0.504	0.000
GS-Linear-Baseline	0.622	0.050	GS-Poly-G	0.233	0.015	OLS-Baseline	0.213	0.009	GS-Linear-GEM	0.439	0.000
GS-Poly-Baseline	0.609	0.050	GS-Poly-Baseline	0.225	0.006	RF	0.224	0.009	GS-Linear-G	0.433	0.000
GS-RBF-Baseline	0.618	0.050	GS-RBF-G	0.231	0.006	GS-RBF-G	0.355	0.009	RF	0.461	0.000
RW	0.630	0.050	OLS-Baseline	0.225	0.001	GS-Linear-Baseline	0.251	0.000	GS-Poly-G	0.445	0.000
GS-Poly-G	0.600	0.050	GS-Poly-GEM	0.376	0.001	GS-Poly-GEM	0.531	0.000	OLS-GEM	0.462	0.000
RF	0.708	0.050	GS-Linear-G	0.242	0.001	GS-RBF-Baseline	0.276	0.000	GS-Poly-GEM	0.617	0.000
GS-Linear-G	0.692	0.050	GS-Linear-GEM	0.502	0.000	GS-Linear-GEM	0.660	0.000	OLS-G	0.483	0.000

[a] For each country, the first column gives the model name, and the second column reports forecast RMSE, scaled by x100. For example, a value of 0.460 corresponds to $RMSE = 0.0046$, or 46 basis points. The third column reports MCS p-values for a model at hand, with $p \geq 0.350$, $p \geq 0.600$, and $p \geq 0.800$ identifying the forecasts in $\widehat{\mathcal{M}}^{*}_{65\%}$, $\widehat{\mathcal{M}}^{*}_{40\%}$ and $\widehat{\mathcal{M}}^{*}_{20\%}$, respectively. The forecast evaluation period is July, 30, 2013, to December 31, 2015 (127 forecasts).

[b] Model names characterize the estimation method (MSNTO or GS for SVR, RF, OLS, or RW), the kernel (Linear, RBF, or Poly) and the vector of economic predictors (Baseline, G, or GEM). The trailing number provides a unique identifier for each tuning parameter setting generated in the MSNTO optimization process, ordered by forecast RMSE.

Table 4
Predictive ability of MSNTO-based SVR forecast combinations[a,b].

Brazil			Mexico			Philippines			Turkey		
Model	RMSE	P_{MCS}	Model	RMSE	P_{MCS}	Model	RMSE	P_{MCS}	Model	RMSE	P_{MCS}
MSNTO-Median-$\widehat{\mathcal{M}}^*_{10\%}$	0.585	1.000	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{60\%}$	0.167	1.000	MSNTO-Median-$\widehat{\mathcal{M}}^*_{10\%}$	0.162	1.000	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{80\%}$	0.283	1.000
MSNTO-Mean-$\widehat{\mathcal{M}}^*_{10\%}$	0.585	0.868	MSNTO-Median-$\widehat{\mathcal{M}}^*_{60\%}$	0.167	0.898	MSNTO-Median-$\widehat{\mathcal{M}}^*_{20\%}$	0.163	0.418	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{60\%}$	0.287	0.359
MSNTO-Mean-$\widehat{\mathcal{M}}^*_{60\%}$	0.589	0.774	MSNTO-Median-$\widehat{\mathcal{M}}^*_{40\%}$	0.168	0.898	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{10\%}$	0.164	0.418	MSNTO-Median-$\widehat{\mathcal{M}}^*_{80\%}$	0.288	0.004
MSNTO-Mean-$\widehat{\mathcal{M}}^*_{40\%}$	0.590	0.774	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{80\%}$	0.169	0.898	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{20\%}$	0.165	0.391	GS-RBF-GEM	0.290	0.004
OLS-GEM	0.594	0.774	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{40\%}$	0.168	0.745	RW	0.171	0.391	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{40\%}$	0.292	0.004
MSNTO-Mean-$\widehat{\mathcal{M}}^*_{80\%}$	0.591	0.352	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{90\%}$	0.171	0.745	MSNTO-Median-$\widehat{\mathcal{M}}^*_{60\%}$	0.170	0.391	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{90\%}$	0.291	0.004
MSNTO-Mean-$\widehat{\mathcal{M}}^*_{20\%}$	0.593	0.352	MSNTO-Median-$\widehat{\mathcal{M}}^*_{90\%}$	0.171	0.624	MSNTO-Median-$\widehat{\mathcal{M}}^*_{40\%}$	0.171	0.391	MSNTO-Median-$\widehat{\mathcal{M}}^*_{60\%}$	0.292	0.001
MSNTO-Median-$\widehat{\mathcal{M}}^*_{80\%}$	0.599	0.136	MSNTO-Median-$\widehat{\mathcal{M}}^*_{80\%}$	0.170	0.624	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{40\%}$	0.170	0.391	MSNTO-Median-$\widehat{\mathcal{M}}^*_{90\%}$	0.292	0.001
MSNTO-Median-$\widehat{\mathcal{M}}^*_{60\%}$	0.600	0.136	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{20\%}$	0.171	0.416	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{60\%}$	0.174	0.277	GS-Linear-Baseline	0.293	0.001
MSNTO-Median-$\widehat{\mathcal{M}}^*_{40\%}$	0.600	0.136	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{10\%}$	0.172	0.394	GS-RBF-GEM	0.187	0.277	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{10\%}$	0.295	0.001
MSNTO-Mean-$\widehat{\mathcal{M}}^*_{90\%}$	0.602	0.136	MSNTO-Median-$\widehat{\mathcal{M}}^*_{20\%}$	0.172	0.062	GS-Poly-Baseline	0.176	0.277	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{20\%}$	0.295	0.001
MSNTO-Median-$\widehat{\mathcal{M}}^*_{90\%}$	0.603	0.136	MSNTO-Median-$\widehat{\mathcal{M}}^*_{10\%}$	0.173	0.062	RF	0.182	0.243	OLS-Baseline	0.314	0.001
OLS-G	0.629	0.136	RF	0.185	0.062	GS-Linear-G	0.186	0.243	MSNTO-Median-$\widehat{\mathcal{M}}^*_{40\%}$	0.299	0.001
MSNTO-Median-$\widehat{\mathcal{M}}^*_{20\%}$	0.605	0.136	GS-RBF-GEM	0.241	0.062	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{90\%}$	0.193	0.243	OLS-G	0.474	0.000
GS-RBF-G	0.637	0.136	GS-RBF-Baseline	0.213	0.062	MSNTO-Median-$\widehat{\mathcal{M}}^*_{90\%}$	0.195	0.243	OLS-GEM	0.472	0.000
GS-Poly-GEM	0.635	0.136	OLS-G	0.230	0.062	MSNTO-Median-$\widehat{\mathcal{M}}^*_{80\%}$	0.197	0.243	GS-Poly-GEM	0.668	0.000
GS-Linear-GEM	0.647	0.136	OLS-GEM	0.221	0.062	GS-Poly-G	0.199	0.243	GS-Linear-G	0.413	0.000
OLS-Baseline	0.667	0.136	OLS-Baseline	0.214	0.019	MSNTO-Mean-$\widehat{\mathcal{M}}^*_{80\%}$	0.196	0.243	GS-Poly-G	0.425	0.000
RW	0.759	0.136	GS-Linear-Baseline	0.236	0.019	GS-Linear-GEM	0.761	0.000	GS-RBF-G	0.413	0.000
GS-Poly-G	0.735	0.136	GS-RBF-G	0.227	0.019	GS-Poly-GEM	0.476	0.000	RW	0.345	0.000
GS-RBF-GEM	0.945	0.136	GS-Poly-G	0.233	0.019	GS-RBF-Baseline	0.264	0.000	GS-RBF-Baseline	0.352	0.000
GS-Poly-Baseline	0.756	0.109	GS-Poly-Baseline	0.225	0.003	GS-Linear-Baseline	0.247	0.000	GS-Linear-GEM	0.354	0.000
GS-RBF-Baseline	0.765	0.109	GS-Poly-GEM	0.316	0.003	GS-RBF-G	0.280	0.000	RF	0.310	0.000
GS-Linear-Baseline	0.775	0.109	RW	0.228	0.003	OLS-G	0.228	0.000	GS-Poly-Baseline	0.322	0.000
RF	0.877	0.109	GS-Linear-G	0.234	0.003	OLS-GEM	0.234	0.000	MSNTO-Median-$\widehat{\mathcal{M}}^*_{10\%}$	0.302	0.000
GS-Linear-G	0.867	0.077	GS-Linear-GEM	0.448	0.000	OLS-Baseline	0.190	0.000	MSNTO-Median-$\widehat{\mathcal{M}}^*_{20\%}$	0.301	0.000

[a] For each country, the first column gives the model name, and the second column reports forecast RMSE, scaled by $\times 100$. The third column reports MCS p-values for a model at hand, with $p \geq 0.350$, $p \geq 0.600$, and $p \geq 0.800$ identifying the forecasts in $\widehat{\mathcal{M}}^*_{65\%}$, $\widehat{\mathcal{M}}^*_{40\%}$, and $\widehat{\mathcal{M}}^*_{20\%}$, respectively. The forecast evaluation period is October, 20, 2014 to December 31, 2015 (62 forecasts).

[b] Model names characterize the estimation method (MSNTO or GS for SVR, RF, OLS or RW), the kernel (Linear, RBF, or Poly) and the the vector of economic predictors (Baseline, G or GEM). MSNTO-Mean-$\widehat{\mathcal{M}}^*_{1-\alpha}$ and MSNTO-Median-$\widehat{\mathcal{M}}^*_{1-\alpha}$ refer to the mean and median forecast combination based on the model confidence set at level α, respectively.

13.6% and 16.9% reduction in forecast RMSE relative to the country-specific best-performing standard SVR and OLS benchmarks, respectively.[27] The respective reduction in forecast RMSE relative to the canonical RW and RF benchmark forecasts is similarly sizable at 18.1% and 15.6%, respectively, on average across countries.

Furthermore, aligned with findings in the previous literature, the mean forecast combinations consistently deliver superior predictive accuracy, entering $\widehat{\mathcal{M}}^*_{65\%}$ most accurate forecasts for all countries; see for example Smith and Wallis (2009), Claeskens et al. (2016), and Wang et al. (2023). At the same time, the predictive performance of the median combinations is more varied across countries. For example, when most median forecast combinations, with the exception of MSNTO-Median-$\widehat{\mathcal{M}}^*_{10\%}$ and MSNTO-Median-$\widehat{\mathcal{M}}^*_{20\%}$, enter $\widehat{\mathcal{M}}^*_{65\%}$ for Mexico, none of the median forecast combinations make it into the most accurate model sets for Turkey. Accordingly, we focus on the mean forecast combinations in our subsequent analysis.

The results in Table 4 further indicate that the predictive accuracy of the mean forecast combinations varies with the MCS test confidence level α employed during model selection, confirming similar findings in Samuels and Sekkel (2017). However, in contrast to these authors' observation that using the highest in their study confidence level, $\alpha = 0.1$, leads to the most accurate combination forecasts in their U.S. macroeconomy-based application, the combination forecasts based on lower confidence levels, namely, at $\alpha = 0.2$ to 0.9, deliver more accurate forecasts for countries in our EME bond market application. Interestingly, MSNTO-Mean-$\widehat{\mathcal{M}}^*_{90\%}$, based on $\alpha = 0.1$, almost never performs well in our EME credit spread forecasts, with Mexico the only country where this model set delivers competitive forecasts, on par with those from MSNTO-Mean-$\widehat{\mathcal{M}}^*_{40\%}$.

Our finding of generally lower MCS test confidence levels delivering competitive combination forecasts relative to those in Samuels and Sekkel (2017) can be explained by higher risk associated with the EME bond markets in our application, contributing to higher variability of the EME credit spreads, with greater market frictions also likely of importance for some markets. In particular, using lower confidence levels, leading to a broader model selection, improves the forecast combination ability to produce accurate forecasts likely by avoiding overfitting in our EME market context. This can also, at least in part, explain the cross-country differences in the α values that generate the most accurate combination forecasts, with the lower confidence levels benefiting borrowers experiencing a deterioration in credit quality, accompanied by heightened credit spread volatility. For example, using relatively high confidence levels, at $\alpha = 0.2$ and 0.4, produces the most accurate mean combination forecasts for Mexico and Turkey whose credit ratings remained unchanged.[28] At the same time, for Brazil, which was downgraded from investment-grade BBB to a speculative grade BB by the S&P during our forecast evaluation period (September 9, 2015), a notably lower confidence level, at $\alpha = 0.9$, delivers the highest predictive accuracy. The change in credit quality is also reflected in heightened volatility of Brazil credit spreads which at 83 basis points is 2.5- to 6-fold higher than 13, 18, and 33 basis points volatility for the Philippines, Mexico, and Turkey, respectively, during our forecast evaluation period. The Philippines is another country that has benefited from using $\alpha = 0.9$, even though, unlike Brazil, its credit rating and credit spreads have remained relatively stable. The underlying reasons are likely to be different for the Philippines and related to higher market frictions associated with its much smaller Eurobond market size, reflecting the smaller size of its economy compared to the other three markets.[29] This is because a relatively small market size is typically associated with higher trading and information costs, due to lower trading volumes and less market liquidity that in turn can hinder the link between macroeconomic predictors and credit spreads, and the credit spread predictability (Audzeyeva & Fuertes, 2018).

5.3. Further empirical observations

Table 3 further shows that there is no persuasive evidence for singling out a particular kernel function as most suited for modeling sovereign credit spreads, with all three kernels employed in the best model sets of various countries. Nevertheless, country-specific evidence suggests that the RBF is a clear preferred kernel for Brazil and Turkey, with all models in $\widehat{\mathcal{M}}^*_{65\%}$ being RBF-based. The two countries' credit quality dynamics differ from those of the other countries in that they experienced a deterioration in their expected intermediate-term credit quality as signaled by a series of negative rating-outlook announcements by the S&P during the forecast evaluation period in Table 3, followed by the rating downgrade to the speculative-grade BB-rating for Brazil (but not for Turkey, whose credit quality had subsequently stabilized). At the same time, Poly is the preferred kernel for Mexico, and both Poly and Linear kernels were employed in the competitive models for the Philippines, the remaining two countries whose credit quality remained stable.

Moreover, our EME-based evidence complements the findings of sovereign credit spread predictability using SVR in Da Silva Fernandes et al. (2019) for the eurozone countries. Furthermore, beyond their findings of the predictive content in the country-specific credit spread data for day- and week-ahead predictive horizons, our evidence suggests that the useful predictive content for quarter-ahead predictions varies for countries with different dynamics in credit quality. Namely, we find that country-specific credit-spread-curve data suffice for predicting credit spreads for Mexico and the Philippines, the two countries with stable credit quality, while exploiting additional predictive content in both the global

[27] The reduction in RMSE $= 1 - \frac{RMSE_{MSNTO-Mean(Median)-\widehat{\mathcal{M}}^*_{1-\alpha}}}{RMSE_{Benchmark}}$ for the mean (median) forecast combination based on the model confidence set at level α.

[28] Even though Turkey was issued several negative outlooks by the S&P between April 3, 2008, and February 7, 2014, its credit rating remained unchanged, with its credit spreads also stabilizing during our forecast evaluation period.

[29] For example, externally financed new bond issuance over the four year period from 2010 to 2013 in total (in billions of US$) was $168.9, $119.2, and $53.5 for Brazil, Mexico, and Turkey, respectively, and only $18.0 billion, for the Philippines IMF, 2014.

and domestic macroeconomic variables improves forecast accuracy for Brazil and Turkey, both experiencing a deterioration in credit quality. While contrasting with findings in Audzeyeva and Fuertes (2018) based on linear predictive models refuting the rational expectation theory of Sargent (1973) and Roll (1970), our evidence is aligned with Diebold et al. (2006), employing the dynamic latent factor model for forecasting U.S. Treasury yields, who document that the expectations hypothesis holds reasonably well during certain periods.

6. Conclusions

Motivated by the well-documented sensitivity of the accuracy of SVR forecasts to kernel function parameter values and the lack of reliable guidance on how to identify these parameters, our study contributes to the SVR forecasting literature by proposing a novel approach for determining SVR kernel parameter values. In contrast to existing approaches aiming to select a single 'best' kernel setting that determines the 'best' SVR predictive model, an impractical goal in many financial market applications, our approach produces a collection of competitive tuning parameter values that generate a set of promising SVR model candidates. We manage this model multiplicity by applying the MCS test to select a subset of the most accurate models. Depending on the modeler's objectives, the resulting forecasts can be employed directly or can serve as inputs in forecast combinations.

In our empirical application, we employed our approach to forecasting the bond credit spreads of four mature emerging market sovereign borrowers. Our study is the first to focus on the application of SVR techniques to forecasting EME credit spreads, to our best knowledge, expanding the nascent SVR literature on sovereign credit spread OOS predictability that to date has focused on developed economies. The evaluation of the accuracy of one-quarter-ahead OOS SVR forecasts that our approach generates, using various kernel functions and predictor sets motivated by economic theory, provides clear evidence that our approach can identify a relatively small set of SVR models with superior OOS forecasting ability. Further, we found that the mean forecast combinations using our most accurate model sets delivered substantial gains in forecast accuracy in both economic and statistical terms relative to RF, standard SVR specifications, and conventional benchmark model forecasts utilized in the credit spreads literature.

Furthermore, our predictability analysis provides evidence that our approach delivers competitive credit spread forecasts for countries with diverse dynamics in credit quality, consistently outperforming the benchmarks and pointing to a better suitability of using an SVR specification for forecasting EME credit spreads relative to linear models typically employed in EME studies (Audzeyeva & Fuertes, 2018; Wang et al., 2019). The superior predictive accuracy of flexible SVR specifications is likely linked to their better ability to capture complex international investor expectations reflected in sovereign EME credit spreads. Moreover, our finding that useful predictor sets differ for countries with various credit quality dynamics suggests that, beyond gains in forecast accuracy over standard techniques, our approach can help provide insights into credit risk pricing for borrowers with diverse dynamics in credit quality.

By delivering competitive forecasts for various countries, our empirical evidence of each of the three kernel functions employed in our analysis corroborates a finding in Sermpinis, Stasinakis, Rosillo, et al. (2017) for European stock market ETFs, countering a widely held belief that the RBF kernel is the optimal choice for modeling financial market series. These findings call for further research into SVR kernel function selection for financial market pricing applications.

Data and code sharing

The Mathematica and Julia codes used for implementing our forecasting approach and for obtaining the results as in Tables 3 and 4 will be accessible at: https://github.com/cemar-llc/SVRForecasts. Since the dataset used in this paper cannot be shared due to the restrictions imposed by the data providers, an alternative "toy", dataset will be made available that can be used in conjunction with our codes. For further inquiries please contact the corresponding author.

CRediT authorship contribution statement

Gary S. Anderson: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alena Audzeyeva:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Much of the work took place at the Federal Reserve Board of Governors where Gary S. Anderson was a Senior Economist. However, the analysis and conclusions in this paper are those of the authors and do not indicate concurrence by members of the research staff or the Federal Reserve Board of Governors. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Data variables: Summary statistics

See Table A.1.

Table A.1
Summary statistics of emerging market credit spreads and predictor variables.

Country Variable	Mean	StDev	Min	Max	AR(1)
US					
Yield curve level	0.043	0.008	0.027	0.058	0.986
Yield curve slope	-0.028	0.013	-0.053	-0.004	0.981
Yield curve curvature	-0.079	0.025	-0.139	-0.022	0.978
Short-term rate volatility	0.000	0.001	0.000	0.004	0.977
Brazil					
Credit spread	0.016	0.008	0.004	0.050	0.973
Credit-spread-curve level	0.025	0.010	0.012	0.061	0.974
Credit-spread-curve slope	-0.011	0.020	-0.061	0.062	0.887
Credit-spread-curve curvature	-0.025	0.040	-0.154	0.069	0.925
Financial risk rating	39.196	3.146	32.500	45.500	0.978
Trade balance	0.067	0.081	-0.159	0.379	0.978
Trade balance volatility	0.054	0.027	0.021	0.157	0.990
Terms-of-trade growth	0.052	8.790	-13.276	19.361	0.997
Terms-of-trade growth volatility	2.747	1.560	0.389	8.899	0.991
Mexico					
Credit spread	0.014	0.006	0.006	0.047	0.972
Credit-spread-curve level	0.021	0.006	0.013	0.044	0.930
Credit-spread-curve slope	-0.009	0.014	-0.044	0.024	0.874
Credit-spread-curve curvature	-0.021	0.032	-0.086	0.063	0.930
Financial risk rating	40.289	1.908	35.500	43.000	0.978
Trade balance	-0.033	0.071	-0.272	0.085	0.958
Trade balance volatility	0.066	0.032	0.024	0.135	0.992
Terms-of-trade growth	-2.778	9.850	-22.243	19.394	0.998
Terms-of-trade growth volatility	3.609	2.973	0.356	14.099	0.994
Philippines					
Credit spread	0.016	0.010	0.005	0.067	0.989
Credit-spread-curve level	0.021	0.007	0.008	0.042	0.954
Credit-spread-curve slope	-0.008	0.017	-0.042	0.080	0.913
Credit-spread-curve curvature	-0.009	0.039	-0.151	0.159	0.945
Financial risk rating	43.157	2.267	36.000	46.500	0.991
Trade balance	-0.258	0.184	-0.595	0.293	0.976
Trade balance volatility	0.154	0.060	0.077	0.337	0.988
Turkey					
Credit spread	0.025	0.009	0.013	0.073	0.964
Credit-spread-curve level	0.029	0.008	0.013	0.063	0.961
Credit-spread-curve slope	-0.007	0.017	-0.055	0.061	0.797
Credit-spread-curve curvature	-0.006	0.030	-0.124	0.127	0.796
Financial risk rating	33.740	2.550	27.000	37.500	0.981
Trade balance	-0.816	0.227	-1.253	-0.202	0.993
Trade balance volatility	0.110	0.041	0.042	0.209	0.989
Terms-of-trade growth	0.857	4.350	-8.864	9.667	0.995
Terms-of-trade growth volatility	1.732	1.183	0.241	5.320	0.993

The table reports mean, standard deviation, minimum, maximum, and 1st order autocorrelation for credit spreads and predictor variables. Treasury yields and credit spreads are given in decimals; for example, 0.043 is equivalent to 430 basis points. The weekly data are from Audzeyeva and Fuertes (2018). The sample period covers the post-GFC recovery period from December 2, 2008 to December 29, 2015.

Appendix B. Lagrangian of the loss function, Eq. (2)

The loss function, Eq. (2), has a dual Lagrangian of the form

$$\mathcal{L} = \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}(\xi_i + \xi_i^*) - \sum_{i=1}^{l}(\eta_i\xi_i + \eta_i^*\xi_i^*) - \sum_{i=1}^{l}\upsilon_i(\epsilon + \xi_i - y_i + k(x_i, x; \Upsilon) + w_0) - \sum_{i=1}^{l}\upsilon_i^*(\epsilon + \xi_i^* + y_i - k(x_i, x; \Upsilon) - w_0) \quad \text{(B.1)}$$

Eq. (B.1) constitutes a quadratic programming problem with well-known properties and solution algorithms. The optimization produces a set of nonzero weights υ_i, υ_i^* which identifies a collection of support vectors, influential observations that determine the optimal set of weights. Deleting the other observations and again solving the quadratic programming problem produces the same set of optimal weights. For our calculations, we used the C++ version of **libsvm**, a widely used open-source SVR implementation.

Appendix C. Pseudocode

Input:

- Time series data: $\{(y_t, x_t)\}_{t=1}^{T}$
  - y_t: target variable (e.g. credit spread)
  - x_t: predictor vector (e.g. Nelson–Siegel factors, global and domestic variables)
- Forecast horizon: q (e.g. $q = 13$ weeks)
- Training sample size: T_0 (e.g. first two-thirds of data), OOS evaluation period: $T_1 = T - T_0$
- SVR kernels: $K \in \{K_{\text{Linear}}, K_{\text{Poly}}, K_{\text{RBF}}\}$
- Model specifications: M, (e.g. $M \in \{\text{Baseline}, G, GEM\}$)
- Parameter space for tuning parameters: $\pi = \{\Upsilon \in \mathbb{R}^{\ell} : a \le \Upsilon \le b\}$
- hv-block cross-validation parameters: v (block size) and h (buffer size)
- MSNTO parameters:
  - Initial candidate count: Λ_1
  - Subsequent candidate set size per cluster: Λ_2
  - Clustering fraction: Π
  - Dispersion multiplier: ρ_Φ
  - Contraction factor: σ
  - Recursion limit: N
- Confidence level for the MCS test: α

Output:

- Combined forecast $\hat{y}_{t+q}^{\text{comb}}$ for each forecasting model, consisting of $\{K, M, \Upsilon^{\dagger}\}$ (with performance metrics, e.g. RMSE and MCS model loss)[30]

Step 1: Data partitioning

- Partition into:
  - Training set: $\{(y_t, x_t)\}_{t=1}^{T_0}$
  - OOS evaluation: $\{(y_t, x_t)\}_{t=T_0+1}^{T}$

Step 2: For each country, kernel K, and model M: Tune SVR kernel parameters using MSNTO to minimize the cross-validation error

Step 2a: Define the cross-validation error function

- For any candidate tuning parameter vector Υ, define the hv-block cross- validation metric as

$$\Phi(\Upsilon) = \frac{1}{(n-2v)n_v}\sum_{i=v+1}^{n-v}\left\|y_{(i:v)} - \hat{y}_{(i:v)}(\Upsilon)\right\|^2$$

  where
  - $n = T_0$ (the training sample size)
  - For each i $(v+1 \le i \le i-v)$:
    - $V_i = \{(y_j, x_j) : j = i-v, \ldots, i+v\}$, size $n_v = 2v+1$
    - Exclude h observations on each side of V_i to ensure independence between training and validation
- Train the SVR (with kernel K and specification M) on the corresponding training data and compute the squared error on V_i

Step 2b: Tuning parameter selection via MSNTO

- **Initialization:** Generate initial candidates $U^{(1)} = \Upsilon_1$ distributed uniformly over π (using e.g. a Niederreiter sequence)
- **Repeat**:
  For iterations $k = 1, 2, \ldots, N$:
  (a) Evaluate $\Phi(\Upsilon)$ for each candidate $\Upsilon \in U^{(k)}$
  (b) Estimate the dispersion among the candidate points:
  $$\hat{B}(\rho, \pi) = \max_{1 \le i \le |U^{(k)}|}\ \min_{j \ne i}\|\Upsilon_i - \Upsilon_j\|$$
  (c) Set the clustering threshold $\rho = \rho_\Phi \hat{B}(\rho, \pi)$
  (d) Cluster candidates within distance ρ of each other
  (e) Contract region around each cluster's minimum $\Upsilon_{\min}$ by σ to produce this recursion's candidates: Υ_i^*
  (f) Generate Λ_2 new points per cluster for $U^{(k+1)}$
  (g) Stop after recursion N
- **Output:** To obtain the final candidate set, denote the indexed family of intermediate recursion results as $\{\Upsilon_1^*, \ldots, \Upsilon_N^*\}$ and compute $\Upsilon^{**} = \bigcup_{i=1}^{N}\Upsilon_i^*$ using e.g. Mathematica's *Union* function to eliminate duplicates[31]

[30] See Step 3 for the description of $\Upsilon^{\dagger}$.

Table D.1
Examples of tuning parameter values.

Country	Model[a]	Kernel parameter values[b]
Brazil		
	MSNTO-RBF-GEM-1	(231.5900, 0.0050, 0.6661)
	MSNTO-RBF-GEM-2	(208.4100, 0.0050, 0.6792)
	MSNTO-RBF-GEM-3	(219.7800, 0.0050, 0.6549)
	MSNTO-RBF-GEM-4	(216.9500, 0.0050, 0.6657)
	MSNTO-RBF-GEM-6	(224.8300, 0.0050, 0.6905)
Mexico		
	MSNTO-Poly-Baseline-1	(768.5600, 0.0040, 2.6659, −4.7129, 1)
	MSNTO-Poly-Baseline-2	(1150.7000, 0.0046, 5.4177, −6.8902, 1)
	MSNTO-Poly-Baseline-3	(1112.5000, 0.0045, 4.2435, −5.0987, 1)
	MSNTO-Poly-Baseline-4	(770.3100, 0.0043, 2.3927, −6.6595, 1)
	MSNTO-Poly-Baseline-5	(798.8800, 0.0041, 2.8955, −1.0733, 1)
The Philippines		
	MSNTO-Poly-Baseline-1	(798.6000, 0.0017, 4.8206, 4.1955, 1)
	MSNTO-Poly-Baseline-2	(726.8400, 0.0027, 1.3455, −11.9360, 1)
	MSNTO-Poly-Baseline-3	(171.1100, 0.0016, 6.8870, −17.0940, 1)
	MSNTO-Poly-Baseline-7	(553.7600, 0.0017, 3.4048, −10.5360, 1)
	MSNTO-Poly-Baseline-5	(393.4200, 0.0023, 4.4379, −14.5550, 1)
Turkey		
	MSNTO-RBF-GEM-1	(321.7500, 0.0017, 23.2390)
	MSNTO-RBF-GEM-2	(204.8400, 0.0014, 29.3890)
	MSNTO-RBF-GEM-3	(263.9900, 0.0016, 28.5110)
	MSNTO-RBF-GEM-4	(308.4900, 0.0016, 29.0050)
	MSNTO-RBF-GEM-5	(314.1700, 0.0015, 29.6820)

[a] For each country given in the first column, the second column lists the top five MSNTO-based SVR predictive models from Table 3, and the third column reports the tuning parameter values for a model at hand.
[b] The RBF kernel parameter set is (C, ϵ, ψ), and the polynomial kernel parameter set is $(C, \epsilon, s, g, \gamma)$.

Step 3: Out-of-sample model training and forecast generation

For each candidate $\Upsilon^{\dagger} \in \Upsilon^{**}$:

- Train SVR with (K, M) on the out-of-sample training data using rolling windows
- Generate the q-step-ahead forecast using a rolling window:

$$\hat{y}_{t+q}^{\Upsilon^{\dagger}} = f(x_t; \Upsilon^*, K, M), \quad t \in \{T_0, T_0+1, \ldots, T-q\}$$

Step 4: Model evaluation via the MCS test

- For each candidate model j (indexed over all Υ^* and (K, M) combinations), compute the loss at each forecast time: $L_{j,t} = (y_{t+q} - \hat{y}_{t+q}^{(j)})^2$
- For each pair (j, j^*), define the loss difference: $\Delta_{jj',t} = L_{j,t} - L_{j',t}$
- Compute the sample mean difference: $\mu_{jj'} = E[\Delta_{jj',t}]$
- **MCS procedure:**
  - Initialize[32] the candidate model set M_0
  - Sequentially test the null hypothesis $H_{0,M} : \mu_{jj'} = 0 \ \forall j, j' \in M_0$ using the maximum t-statistic over all pairs.
  - Eliminate the model with the worst significant inferiority (lowest p-value) until the null can no longer be rejected at significance level α
- Let the remaining set be the model confidence set: $\widehat{M}^*_{1-\alpha}$

Step 5: Forecast combination

For each forecast origin t (with a rolling update), let the forecasts from models in $\widehat{M}^*_{1-\alpha}$ be $\{\hat{y}_{t+q}^{(j)}\}$

- **Combined forecast options:**
  - Mean combination: $\hat{y}_{t+q}^{\text{comb}} = \frac{1}{|\widehat{M}^*_{1-\alpha}|} \sum_{j \in \widehat{M}^*_{1-\alpha}} \hat{y}_{t+q}^{(j)}$
  - Median combination: $\hat{y}_{t+q}^{\text{comb}} = \text{median}\{\hat{y}_{t+q}^{(j)} : j \in \widehat{M}^*_{1-\alpha}\}$

Step 6: Rolling forecast updating and benchmarking

- Update the training window (e.g. roll forward by one week) and repeat Steps 3–5 for each model combination
- Compare the performance (e.g. forecast RMSE) of $\hat{y}_{t+q}^{\text{comb}}$ with benchmarks such as standard SVR with grid search, random forests, OLS, and random walk forecasts

Return: The combined forecasts $\{\hat{y}_{t+q}^{\text{comb}}\}$ for each out-of-sample period, along with performance metrics

Appendix D. Tuning parameter values

See Table D.1.

[31] Mathematica's *Union* function considers real numbers equal if they differ in at most their last seven binary digits.
[32] If the number of models is large, a performance metric (e.g. forecast RMSE) can be used as a practical tool for reducing the model set and limiting the computational burden.

Appendix E. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2026.03.004.

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